

Shankar 16.1.3

Trial function of $\psi(x) = e^{-\alpha x/2}$

for attractive delta function potential $V = -aV_0 \delta(x)$

$$E(\alpha) = \frac{\int_{-\infty}^{\infty} e^{-\alpha x/2} \left(-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} - aV_0 \delta(x) \right) e^{-\alpha x/2} dx}{\int_{-\infty}^{\infty} e^{-\alpha x} dx}$$

$$= \frac{-\frac{\hbar^2}{2m} \left(-\frac{1}{2} \sqrt{\alpha} \sqrt{\pi} \right) - aV_0}{\sqrt{\frac{\pi}{\alpha}}}$$

$$E(\alpha) = \frac{\hbar^2}{4m} \alpha - \frac{aV_0}{\sqrt{\pi}} \alpha^{1/2}. \text{ Now minimize energy.}$$

$$0 = \frac{dE}{d\alpha} = \frac{\hbar^2}{4m} - \frac{aV_0}{2\sqrt{\pi}} \frac{1}{\alpha^{1/2}}$$

$$\alpha_0 = \frac{4m^2 a^2 V_0^2}{\hbar^4 \pi} \quad \text{Plug back into } E(\alpha) \text{ to}$$

obtain energy.

$$E(\alpha_0) = \frac{\hbar^2}{4m} \left(\frac{4m^2 a^2 V_0^2}{\hbar^4 \pi} \right) - \frac{aV_0}{\sqrt{\pi}} \left(\frac{2maV_0}{\hbar^2 \sqrt{\pi}} \right)$$

$$E(\alpha_0) = -\frac{a^2 V_0^2 m}{\hbar^2 \pi} > -\frac{a^2 V_0^2 m}{2\hbar^2}$$

$\uparrow$ gaussian estimate energy $\uparrow$ exact bound state energy

Shankar 17.2.1 a) Consider $H' = \lambda x^4$ for the oscillator.
Calculate E_n'

$$E_n' = \langle n | H' | n \rangle$$

Recall $x = \sqrt{\frac{\hbar}{2m\omega}} (a + a^\dagger)$, so $\lambda x^4 = \frac{\lambda \hbar^2}{4m^2\omega^2} (a + a^\dagger)^2 (a + a^\dagger)^2$

Thus $E_n' = \frac{\lambda \hbar^2}{4m^2\omega^2} \langle n | (a + a^\dagger)^2 \cdot (a + a^\dagger)^2 | n \rangle$

Let's split the inner product in two and evaluate:
 $(a + a^\dagger)^2 | n \rangle$ and $\langle n | (a + a^\dagger)^2$ separately...

$$(a + a^\dagger)^2 | n \rangle = \sqrt{n(n-1)} | n-2 \rangle + (2n+1) | n \rangle + \sqrt{(n+1)(n+2)} | n+2 \rangle$$

Similarly,

$$\langle n | (a + a^\dagger)^2 = \sqrt{n(n-1)} \langle n-2 | + (2n+1) \langle n | + \sqrt{(n+1)(n+2)} \langle n+2 |$$

Putting the inner product back together and taking advantage of orthogonality,

$$E_n' = \frac{\lambda \hbar^2}{4m^2\omega^2} (n(n-1) + (2n+1)^2 + (n+1)(n+2))$$

$$E_n' = \frac{3\lambda \hbar^2}{4m^2\omega^2} (1 + 2n + 2n^2) \quad \checkmark$$

b) Recall that $E_n^0 = \hbar\omega(n + \frac{1}{2})$

The zeroth order energy (unperturbed) goes like n^1 .

The first order energy correction goes like n^2 .

Thus, regardless of the smallness of λ , for some large enough n , the higher order energies increase, and perturbation theory fails. Physically, at high n 's, the particle has a higher probability of being found at large x 's so λx^4 is no longer a small perturbation.

Shankar 17.24) Prove Thomas-Reiche-Kuhn sum rule.

$$\sum_n (E_n - E_n) |\langle n' | X | n \rangle|^2 = \frac{\hbar^2}{2m}$$

We write the left hand side as

$$\sum_n (E_n - E_n) \langle n | X | n' \rangle \langle n' | X | n \rangle$$

$$= \sum_n [\langle n | X | n' \rangle \langle n' | X | n \rangle - \langle n | H X | n' \rangle \langle n' | X | n \rangle]$$

Eliminating the complete set of states

$$= \langle n | X H X | n \rangle - \langle n | H X X | n \rangle$$

$$= -\langle n | [H, X] | n \rangle \quad (1)$$

Alternatively, if we switch where we place E_n

$$= \sum_n [\langle n | X | n' \rangle \langle n' | H X | n \rangle - \langle n | X | n' \rangle \langle n' | X H | n \rangle]$$

$$= -\langle n | [X, X H] | n \rangle \quad (2)$$

Since (1) and (2) are equal, their average is nothing but themselves.

$$= -\frac{1}{2} ([H, X] + [X, X H])$$

$$= -\frac{1}{2} ([H, X], X). \text{ We are left to evaluate this commutator.}$$

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$$\text{Shankar 17.2 (H cont)} \quad [H, x] = \left[\frac{p^2}{2m}, x \right] + \cancel{[V(x), x]}^0$$

$$= -\frac{\hbar^2}{m} \frac{d}{dx}$$

and then

$$[[H, x], x] = -\frac{\hbar^2}{m}$$

$$= -\frac{1}{2}([H, x]^{SO}, x) = \frac{\hbar^2}{2m}. \quad \text{QED.}$$

$$4a) \hat{H}_0 = \frac{\hat{p}^2}{2m} + \frac{m\omega^2}{2} \hat{x}^2 \quad \hat{H}' = \kappa \hat{x}^2$$

Define ω' such that $\left(\frac{m\omega^2}{2} + \kappa\right) \equiv \frac{m\omega'^2}{2}$

$$\Rightarrow \omega'^2 = \omega^2 + \frac{2\kappa}{m}$$

$$\omega' = \sqrt{\omega^2 + \frac{2\kappa}{m}}$$

So full Hamiltonian is $\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega'^2 \hat{x}^2$, and this is simply a harmonic oscillator.

$$\text{So } E_n = (n + \frac{1}{2})\hbar\omega'$$

$$4b) E_n^{(1)} = \langle n^{(0)} | \hat{H}' | n^{(0)} \rangle$$

$$= \kappa \langle n^{(0)} | \hat{x}^2 | n^{(0)} \rangle$$

$$= \kappa \frac{\hbar}{2m\omega} \langle n^{(0)} | (\hat{a} + \hat{a}^\dagger)^2 | n^{(0)} \rangle$$

$$= \kappa \frac{\hbar}{2m\omega} \langle n^{(0)} | (\hat{a}\hat{a}^\dagger + \hat{a}^\dagger\hat{a}) | n^{(0)} \rangle$$

$$= \kappa \frac{\hbar}{2m\omega} \left[\langle n^{(0)} | \hat{a}\hat{a}^\dagger | n^{(0)} \rangle + \langle n^{(0)} | \hat{a}^\dagger\hat{a} | n^{(0)} \rangle \right]$$

$$= \kappa \frac{\hbar}{2m\omega} (3n^2 + 2n + 1)$$

$$= \kappa \frac{\hbar}{2m\omega} \quad \text{for ground state } (n=0).$$

Compare to exact:

Expand E_0 to first order in κ :

$$E_0 = \frac{1}{2}\hbar\sqrt{\omega^2 + \frac{2\kappa}{m}} = \frac{1}{2}\hbar\omega \left(1 + \frac{2\kappa}{m\omega^2}\right)^{\frac{1}{2}} \approx \frac{1}{2}\hbar\omega \left(1 + \frac{\kappa}{m\omega^2}\right)$$

$$= E_0^{(0)} + \frac{\kappa\hbar}{2m\omega} = E_0^{(0)} + E_0^{(1)} \quad \text{from perturbation theory. } \checkmark$$

⑤ Liboff 13.4

$$E_2^{(0)} = \langle 2^{(0)} | \hat{H}_0 | 2^{(0)} \rangle = \frac{2\hbar^2\pi^2}{mL^2}$$

$$E_2^{(1)} = \langle 2^{(0)} | \hat{H} | 2^{(0)} \rangle = \int_{-\frac{a}{2}}^{\frac{a}{2}} dx \psi_2^{(0)*}(x) V_0 \psi_2^{(0)}(x) = V_0 \int_{-\frac{a}{2}}^{\frac{a}{2}} dx \left[\frac{2}{L} \sin^2\left(\frac{2\pi x}{L}\right) \right]$$

Take $\xi = \frac{2\pi x}{L}$

$$= \frac{2V_0}{L} \int_{-\frac{a\pi}{L}}^{\frac{a\pi}{L}} \frac{L}{2\pi} \sin^2 \xi d\xi = \frac{V_0}{\pi} \left[\frac{1}{2} \xi - \frac{1}{4} \sin 2\xi \right]_{-\frac{a\pi}{L}}^{\frac{a\pi}{L}}$$

$$= \frac{V_0}{\pi} \left(\frac{a\pi}{L} - \frac{1}{2} \sin \frac{2a\pi}{L} \right)$$

⑥ Small correction if $\frac{a\pi}{L}$ is small.

⑦ even parity (odd n) eigenstates have peaks in their probability density at the origin — i.e. they "see" more of the perturbation than do the odd ~~even~~ parity states (which go to zero in the vicinity of the perturbation.)